Lineweaver–Patel Gate Diagram:

A Cantor-Spectral Address System for 58 Objects

Spanning 61 Orders of Magnitude in Mass

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Research Article

Abstract

Background: Lineweaver and Patel (2023) demonstrated that every known object in the observable universe occupies a triangular region on a log(mass)–log(radius) diagram, bounded by the Schwarzschild line from below at high mass and the Compton line from below at low mass [1]. Separately, the Aubry–André–Harper (AAH) Cantor spectrum has been shown to predict the atomic radius ratios for 54 elements with a 6.7% mean error using zero adjustable parameters [2]. Whether these two frameworks share a common spectral architecture has not been systematically investigated.

Methods: We compiled a catalog of 58 objects with independently measured masses and radii, spanning from electron neutrinos ($m \approx 10^{-37}$ kg) to the observable universe ($m \approx 10^{53}$ kg). Each object was assigned a φ-bracket address $bz = \mathrm{round}[\log(r/l_P)/\log(\varphi)]$ and verified against the boundary condition $\rho \geq \max(1/\mu, 2\mu)$ in Planck units. The effective density slope $\Gamma = \log(\rho)/\log(\mu)$ was computed for every object and compared to the three characteristic values predicted by the gate-angle mapping $\Gamma = (2\Theta - 1)/(2\Theta + 1)$ from the companion atomic-radius formula.

Results: All 58 objects satisfied the boundary conditions. The 11 fundamental particles lie on the Compton line with $\Gamma = -1.000$ exactly. The 8 black holes lie on the Schwarzschild line with mean $|\Gamma - 1| = 0.007$. The 38 atomic-density objects occupied the interior of the triangle. The φ-bracket system mapped the full catalog onto a single integer scale from bz = 81 (top quark) to bz = 294 (observable universe). The product of the total bracket span (N = 294) and the AAH wall fraction (W = 0.4671) yielded $N \times W = 137.3$, which was within 0.2% of the inverse fine-structure constant $\alpha^{-1} = 137.036$. The four-gate transmission probability $W^4 = 0.048$ agreed with the observed cosmic baryon fraction $\Omega_b = 0.049$ to within 3.4%.

Conclusions: The φ-bracket address system, derived from the AAH Cantor spectrum, provides a consistent integer coordinate that locates every tested object from fundamental particles to the observable universe on the Lineweaver–Patel diagram. The quantitative correspondences $N \times W \approx \alpha^{-1}$ and $W^4 \approx \Omega_b$ warrant further investigation as potential structural connections between quasiperiodic spectral theory and cosmological parameters.

Keywords: Lineweaver–Patel diagram; mass–radius relation; Schwarzschild radius; Compton wavelength; Planck units; golden ratio; Aubry–André–Harper model; Cantor spectrum; fine-structure constant; baryon fraction; φ-bracket address

1. Introduction

The observation that every object in the observable universe is bounded by two fundamental lines on a log(mass)–log(radius) plot—the Schwarzschild line ($r = 2Gm/c^2$, slope +1) from below at high mass and the Compton line ($\lambda = \hbar/mc$, slope −1) from below at low mass—was elegantly synthesized by Lineweaver and Patel in their 2023 analysis [1]. The two boundaries are perpendicular in log-log space, intersect near the Planck mass–length scale, and define a triangular allowed region containing all known matter from neutrinos to the observable universe. This diagram has been widely adopted as a pedagogical and analytical tool [3, 4].

Independently, a companion study demonstrated that the Cantor-set energy spectrum of the Aubry–André–Harper (AAH) Hamiltonian at its self-dual critical point encodes quantitative information about atomic radius ratios, predicting r(vdW)/r(cov) for 54 elements with a mean error of 6.7% using zero adjustable parameters [2]. The same framework introduced a four-gate architecture (σ_1–σ_4) in which each gate shares a universal transmission constant $L = 1/\varphi^4$ and a gate angle Θ that parameterizes the periodic-table position.

In this work, we investigate whether the φ-bracket address system derived from the AAH spectrum can serve as a consistent coordinate system for the Lineweaver–Patel diagram. We compile a catalog of 58 objects with independently determined masses and radii, spanning 61 orders of magnitude in mass and 62 in radius, assign each a bracket address, verify the boundary conditions, and test three quantitative predictions of the spectral framework.

2. Methods

2.1 Boundary conditions in Planck units

The Schwarzschild radius $r_s = 2Gm/c^2$ and the Compton wavelength $\lambda_c = \hbar/mc$ define two minimum-size boundaries. In Planck units ($\mu = m/m_P$, $\rho = r/l_P$), these become $\rho \geq 2\mu$ (Schwarzschild: no object smaller than its event horizon) and $\rho \geq 1/\mu$ (Compton: no object more localized than its wavelength). Every physical object must satisfy the following:

$$\boldsymbol{\rho \geq \max(1/\mu, 2\mu)}$$

The fundamental particles satisfy $\rho = 1/\mu$ (on the Compton line), the black holes satisfy $\rho = 2\mu$ (on the Schwarzschild line), and all the other objects lie strictly above both boundaries.

2.2 φ-Bracket address

Each object's radius is mapped to a dimensionless integer via $bz = \mathrm{round}[\log(r/l_P)/\log(\varphi)]$, where $\varphi = (1 + \sqrt{5})/2$ is the golden ratio and $l_P = 1.616 \times 10^{-35}$ m is the Planck length. This maps the continuous radial coordinate onto an integer scale in which the Planck length is bz = 0 and the Hubble radius (4.4×10^{26} m) is bz = 294. The total span N = 294 admits the Zeckendorf decomposition N = 233 + 55 + 5 + 1 = F(13) + F(10) + F(5) + F(2).

2.3 Effective density slope

For each object, the effective slope $\Gamma = \log_{10}(\rho)/\log_{10}(\mu)$ was computed, where ρ and μ are the Planck unit radius and mass, respectively. Objects on the Compton line have $\Gamma = -1$, objects on the Schwarzschild line have $\Gamma = +1$, and objects following constant-density scaling ($m \propto r^3$) have $\Gamma = +1/3$. The gate-angle mapping $\Gamma = (2\Theta - 1)/(2\Theta + 1)$, where Θ is the gate angle from the atomic-radius formula [2], is used to predict these three values at $\Theta \to 0$, $\Theta \to \infty$, and $\Theta = 1$.

2.4 Object catalog

A catalog of 58 objects was compiled from standard references: 11 fundamental particles (PDG 2024 [5]), 6 atoms and nuclei (CODATA 2018, Bondi [6], Alvarez [7]), 6 molecules and biological structures, 4 macroscopic objects, 7 solar system bodies (NASA Planetary Fact Sheets [8]), 6 stars (various; see Table S1), 3 compact objects, 8 black holes (EHT [9], LIGO [10]), 6 galaxies and clusters, and the observable universe (Planck 2018 [11]). The particle radii were set to their Compton wavelengths; the black hole radii were set to their Schwarzschild radii; and all the other radii were from observational data. The complete catalog with sources is provided in Table S1.

2.5 AAH spectral constants

The wall fraction $W = 0.4671$ and gate transmission constant $L = 1/\varphi^4 = 0.14590$ were computed from the 233-site AAH Hamiltonian at $V = 2J$ as described in the companion study [2]. No constants were fitted to the cosmological data presented here.

2.6 Computational tools

All computations were performed in Python 3.12 using NumPy 1.26. Physical constants were taken from CODATA 2018. Large language models (Claude, Anthropic; Grok, xAI) were used for catalog compilation assistance and editorial refinement. All the scientific content, framework design, and conclusions are the sole work of the author.

3. Results

3.1 Boundary verification

All 58 objects in the catalog satisfied the boundary condition $\rho \geq \max(1/\mu, 2\mu)$. The 11 fundamental particles lie exactly on the Compton line ($\rho = 1/\mu$) with zero margins, while being 33–58 log-decades above the Schwarzschild line. The 8 black holes lie exactly on the Schwarzschild line ($\rho = 2\mu$) with zero margins, while being 78–99 log-decades above the Compton line. The 38 atomic-density objects and the observable universe all lay strictly above both boundaries, with minimum margins ranging from 0.3 to 44 log-decades.

3.2 Bracket address distribution

The φ-bracket system mapped the full catalog onto a single integer scale from bz = 81 (top quark, Higgs boson) to bz = 294 (observable universe), spanning 213 brackets. Table 1 summarizes the distribution by object class. Fundamental particles occupied brackets 81–139, atomic-scale objects spanned 98–283, black holes occupied 188–236, and the observable universe sat at the maximum bracket bz = 294. Table 3 presents the bracket addresses and density slopes for 20 representative objects.

Table 1. Summary of boundary verification and density slopes by object class. All 58 objects satisfy the gate inequality.

Object class	N	bz range	Mean Γ	Expected Γ	|Γ| deviation

Fundamental particles	11	81–139	−1.000	−1	0.000
Black holes	8	188–236	+1.007	+1	0.007
Atomic-density	38	98–283	variable	+1/3 (bulk)	see text
Observable universe	1	294	+1.010	≈1	0.010
Total	**58**	**81–294**	—	—	**58/58 in △**

3.3 Density slope classification

The effective density slope Γ provided a clean three-way classification (Table 1). All 11 particles had Γ = −1.000 for numerical precision, confirming their position on the Compton line. The 8 black holes had a mean Γ = +1.007, deviating from the expected value of +1.000 by 0.7%, which is attributable to the factor $\log_{10}(2) = 0.301$ in the Schwarzschild formula ($\rho = 2\mu$ rather than $\rho = \mu$). The observable universe had Γ = +1.010, which is consistent with its position near both boundary lines simultaneously.

The Γ values of the 38 atomic-density objects varied systematically with scale: the atoms and molecules had a large negative Γ (dominated by the Compton-line proximity of their constituent particles), macroscopic objects had Γ values ranging from 1.1 to 3.7, and astrophysical objects converged toward Γ ≈ 1.1–1.2 (approaching the Schwarzschild line as gravitational self-energy becomes significant).

3.4 Quantitative correspondences

Three quantitative correspondences between the AAH spectral constants and independently measured quantities were tested (Table 2).

First, the product N × W = 294 × 0.4671 = 137.3 agreed with the inverse fine-structure constant $\alpha^{-1} = 137.036$ to within 0.2%. If interpreted as the total number of Cantor wall boundaries spanning the allowed triangle, this implies one electromagnetic coupling unit per 2.14 brackets.

Second, the four-gate transmission probability $W^4 = (0.4671)^4 = 0.0476$ agreed with Planck Collaboration's cosmic baryon fraction $\Omega_b = 0.0493 \pm 0.0003$ [11] to within 3.4%. No cosmological data were used in deriving W.

Third, the gate-angle interpolator $\Gamma = (2\Theta - 1)/(2\Theta + 1)$ reproduced the three characteristic slopes exactly: $\Gamma(\Theta \to 0) = -1$ (Compton), $\Gamma(\Theta = 1) = +1/3$ (atomic density), and $\Gamma(\Theta \to \infty) = +1$ (Schwarzschild).

Table 2. Quantitative correspondences between AAH spectral constants and physical observables.

Quantity	Predicted	Observed	Deviation
$N \times W = \alpha^{-1}$	137.3	137.036	0.2%
$W^4 = \Omega(b)$	0.0476	0.0493 ± 0.0003	3.4%
Γ(particles) on Compton	−1.000	−1.000	exact
Γ(BH) on Schwarzschild	+1.000	+1.007	0.7%
Γ(Θ=1) = atomic density	+1/3	+1/3	exact

Table 3. Bracket addresses and density slopes for 20 representative objects. Margin = minimum log-decades above either boundary (0.0 = on the line).

Object	log μ	log ρ	bz	Γ	Class	Margin
Electron neutrino	-29.1	29.1	139	-1.000	particle	0.0
Electron	-22.4	22.4	107	-1.000	particle	0.0
Proton	-19.1	19.1	91	-1.000	particle	0.0
Top quark	-16.9	16.9	81	-1.000	particle	0.0
Hydrogen atom	-19.1	24.9	119	-1.301	atomic	5.8
Uranium-235 nucleus	-16.8	20.6	98	-1.228	atomic	3.8
Bacterium (*E. coli*)	-7.3	28.8	138	-3.924	atomic	21.4
Human (70 kg)	9.5	34.8	166	3.655	atomic	24.9
Moon	30.5	41.0	196	1.344	atomic	10.2
Earth	32.4	41.6	199	1.282	atomic	8.9
Jupiter	34.9	42.6	204	1.220	atomic	7.4
Sun	38.0	43.6	209	1.149	atomic	5.4
Betelgeuse	39.2	46.6	223	1.189	atomic	7.1
White dwarf (Sirius B)	38.0	41.5	199	1.094	atomic	3.3
Neutron star (typical)	38.1	38.8	186	1.019	atomic	0.4
Neutron star (massive)	38.3	38.9	186	1.016	atomic	0.3
Stellar BH (10 M☉)	39.0	39.3	188	1.008	bh	0.0
Stellar BH (30 M☉)	39.4	39.7	190	1.008	bh	0.0
Stellar BH (62 M☉)	39.8	40.0	192	1.008	bh	0.0
Sgr A*	44.6	44.9	215	1.007	bh	0.0
M87*	47.8	48.1	230	1.006	bh	0.0
Milky Way (visible)	49.3	55.5	266	1.126	atomic	5.9
Milky Way (halo)	50.1	56.3	269	1.122	atomic	5.8
Observable universe	60.8	61.4	294	1.010	universe	0.3

4. Discussion

4.1 The bracket address as a universal coordinate

The φ-bracket system assigns every object in the catalog a single integer that encodes its radial scale relative to the Planck length. The golden ratio base ensures that successive brackets are spaced by a factor of $\varphi \approx 1.618$, and the total span $N = 294$ decomposes into Fibonacci numbers reflecting the self-similar structure of the underlying AAH Cantor spectrum [2, 12, 13]. The bracket address is fully determined by the object's radius and requires no knowledge of its mass, composition, or internal structure.

4.2 Connection to the atomic-radius formula

The same four-gate architecture (σ_1–σ_4) and gate angle Θ derived in the companion study [2]—which predicts r(vdW)/r(cov) for 54 elements with 6.7% mean error—maps onto the Lineweaver–Patel triangle via the slope Γ(Θ). At the atomic scale, Θ determines the ratio of two radii; at the cosmological scale, it determines the position of the object within the allowed region. The gate transmission constant $L = 1/\varphi^4$ is the same at both scales.

4.3 Interpretation of the baryon fraction

The correspondence $W^4 \approx \Omega_b$ admits a physical interpretation within the gate framework: each of the four Cantor gates transmits a fraction W of the total energy, and baryonic matter is the residue that crosses all four boundaries. The three complementary terms in the partition identity $1/\varphi + 1/\varphi^3 + 1/\varphi^4 = 1$ (values 0.618, 0.236, 0.146) are numerically close to current estimates for dark energy, dark matter, and baryonic fractions [11]. The agreement of $W^4 = 0.048$

with $\Omega_b = 0.049$ to within 3.4% is notable given that W was derived entirely from 233-site AAH diagonalization with no cosmological input.

4.4 The factor of 2 and the Planck intersection

Lineweaver and Patel note that the Schwarzschild radius of a Planck-mass object is $r_s = 2l_P$, not l_P—the two boundaries do not intersect exactly at the Planck point but are offset by a factor of 2 [1]. Within the spectral framework, this factor corresponds to the two orthogonal spectral axes (gold and silver) required for baryonic confinement: $r_s = Gm/c^2 + Gm/c^2$, where each term represents one fold axis. A maximally rotating Kerr black hole, whose horizons merge at $r = Gm/c^2$, corresponds to a single fold plane.

4.5 Limitations

Several important limitations should be noted. The $\Gamma(\Theta)$ mapping is a mathematical identification rather than a derived physical relationship; a rigorous derivation connecting the atomic Pythagorean geometry to cosmological density scaling remains to be established. The agreements $N \times W \approx \alpha^{-1}$ and $W^4 \approx \Omega_b$ could be high-precision numerical coincidences rather than evidence of structural connection. The factor-of-2 interpretation, while geometrically suggestive, is not independently testable. The effective slope Γ for atomic-scale objects does not converge to the predicted +1/3 because these objects are dominated by quantum rather than gravitational self-energy, placing them far above both boundary lines where Γ is poorly constrained. This report quantifies these correspondences for further investigation rather than claiming a derivation of cosmological parameters from the AAH spectrum.

5. Conclusions

We compiled a catalog of 58 objects spanning 61 orders of magnitude in mass and verified that all satisfy the Lineweaver–Patel boundary conditions when expressed in Planck units. The φ-bracket address system, derived from the AAH Cantor spectrum, assigns each object a consistent integer coordinate from bz = 81 (top quark) to bz = 294 (observable universe). Fundamental particles lie exactly on the Compton line ($\Gamma = -1$), black holes on the Schwarzschild line ($\Gamma \approx +1$), and all other objects within the allowed triangle.

The product $N \times W = 137.3 \approx \alpha^{-1}$ and the four-gate probability $W^4 = 0.048 \approx \Omega_b$ represent quantitative correspondences between the AAH spectral framework and independently measured physical constants, computed with zero free parameters and no cosmological input. Whether these reflect a deep structural connection or are numerical coincidences remains an open question that may be addressable through further computational and observational work.

Declarations

Competing Interests

The author is the founder of iBuilt Ltd. Patent application No. 19/560,637 and additional provisional patents have been filed through iBuilt Ltd.

Funding

No external funding.

Data Availability

The complete 58-object catalog with all computed quantities is provided in Table S1. Source code: https://github.com/thusmann5327/Unified_Theory_Physics.

Use of AI-assisted Tools

Large language models (Claude, Anthropic; Grok, xAI) were used for catalog compilation, numerical verification, and editorial refinement. All the scientific content, framework design, and conclusions are the sole work of the author.

Ethics Approval

Not applicable.

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Table S1. Complete 58-Object Catalog

All masses and radii from published sources are cited. $\mu = m/m_P$, $\rho = r/l_P$ in Planck units. $bz = \text{round}[\log(\rho)/\log(\varphi)]$. $\Gamma = \log(\rho)/\log(\mu)$. All 58 objects satisfy $\rho \geq \max(1/\mu, 2\mu)$.

Object	log μ	log ρ	bz	Γ	Class	Source
Electron neutrino	-29.09	29.09	139	-1.000	particle	PDG 2024 (0.1 eV upper bound)
Electron	-22.38	22.38	107	-1.000	particle	CODATA 2018
Muon	-20.06	20.06	96	-1.000	particle	PDG 2024
Pion (charged)	-19.94	19.94	95	-1.000	particle	PDG 2024
Proton	-19.11	19.11	91	-1.000	particle	CODATA 2018
Neutron	-19.11	19.11	91	-1.000	particle	CODATA 2018
Tau lepton	-18.84	18.84	90	-1.000	particle	PDG 2024
W boson	-17.18	17.18	82	-1.000	particle	PDG 2024
Z boson	-17.13	17.13	82	-1.000	particle	PDG 2024
Higgs boson	-16.99	16.99	81	-1.000	particle	PDG 2024
Top quark	-16.85	16.85	81	-1.000	particle	PDG 2024
Hydrogen atom	-19.11	24.87	119	-1.301	atomic	Bohr radius × BASE
Helium atom	-18.52	24.94	119	-1.347	atomic	vdW radius, Bondi 1964
Carbon atom	-18.04	25.02	120	-1.387	atomic	vdW radius, Bondi 1964
Iron atom	-17.37	25.08	120	-1.444	atomic	vdW radius, Alvarez 2013
Uranium atom	-16.74	25.06	120	-1.497	atomic	vdW radius, Bondi 1964
Uranium-235 nucleus	-16.75	20.56	98	-1.228	atomic	Charge radius, Angeli 2013
Water molecule	-17.86	25.08	120	-1.404	atomic	vdW radius
Hemoglobin	-14.31	26.30	126	-1.838	atomic	Perutz 1960
DNA molecule (1 Mbp)	-10.32	25.79	123	-2.498	atomic	Typical chromatin fiber
Bacterium (*E. coli*)	-7.34	28.79	138	-3.924	atomic	Typical *E. coli*
Red blood cell	-5.91	29.39	141	-4.977	atomic	Typical erythrocyte
Grain of sand	0.48	31.49	151	66.272	atomic	Typical quartz grain
Human (70 kg)	9.51	34.75	166	3.655	atomic	Typical adult
Blue whale	12.84	35.97	172	2.802	atomic	Typical adult
Great Pyramid of Giza	17.44	36.85	176	2.113	atomic	Height as effective radius
Mount Everest	22.57	38.43	184	1.703	atomic	Approximate cone
Ceres	28.63	40.47	194	1.413	atomic	NASA/Dawn
Moon	30.53	41.03	196	1.344	atomic	NASA fact sheet
Mars	31.47	41.32	198	1.313	atomic	NASA fact sheet
Earth	32.44	41.60	199	1.282	atomic	NASA fact sheet
Neptune	33.67	42.18	202	1.253	atomic	NASA fact sheet
Saturn	34.42	42.56	204	1.237	atomic	NASA fact sheet
Jupiter	34.94	42.64	204	1.220	atomic	NASA fact sheet
Proxima Centauri	37.05	42.82	205	1.156	atomic	Boyajian 2012
Sun	37.96	43.63	209	1.149	atomic	IAU nominal values

Sirius A	38.28	43.87	210	1.146	atomic	Liebert 2005
Vega	38.29	44.01	211	1.149	atomic	Aufdenberg 2006
Betelgeuse	39.18	46.58	223	1.189	atomic	Joyce 2020
UY Scuti	38.81	46.87	224	1.208	atomic	Arroyo-Torres 2013
White dwarf (Sirius B)	37.97	41.55	199	1.094	atomic	Barstow 2005
Neutron star (typical)	38.11	38.83	186	1.019	atomic	Ozel & Freire 2016
Neutron star (massive)	38.29	38.91	186	1.016	atomic	PSR J0740+6620, Cromartie 2020
Stellar BH (10 M☉)	38.96	39.26	188	1.008	bh	Theoretical (Schwarzschild)
Stellar BH (30 M☉)	39.44	39.74	190	1.008	bh	LIGO typical
Stellar BH (62 M☉)	39.75	40.05	192	1.008	bh	GW150914 remnant
IMBH (1000 M☉)	40.96	41.26	197	1.007	bh	Typical IMBH
Sgr A*	44.58	44.88	215	1.007	bh	EHT/GRAVITY, Do 2019
M87*	47.77	48.07	230	1.006	bh	EHT Collaboration 2019
TON 618	48.78	49.08	235	1.006	bh	Shemmer 2004
Phoenix A	48.96	49.26	236	1.006	bh	Hlavacek-Larrondo 2022
Milky Way (visible)	49.26	55.49	266	1.126	atomic	Bland-Hawthorn & Gerhard 2016
Milky Way (halo)	50.14	56.27	269	1.122	atomic	Bland-Hawthorn & Gerhard 2016
Andromeda (M31)	50.05	55.62	266	1.111	atomic	Kafle 2018
Virgo Cluster	53.04	57.48	275	1.084	atomic	McLaughlin 1999
Coma Cluster	52.81	57.58	276	1.090	atomic	Kubo 2007
Laniakea Supercluster	54.96	59.17	283	1.077	atomic	Tully 2014
Observable universe	60.84	61.43	294	1.010	universe	Planck 2018 + Hubble radius

Figure 1 and Table S1

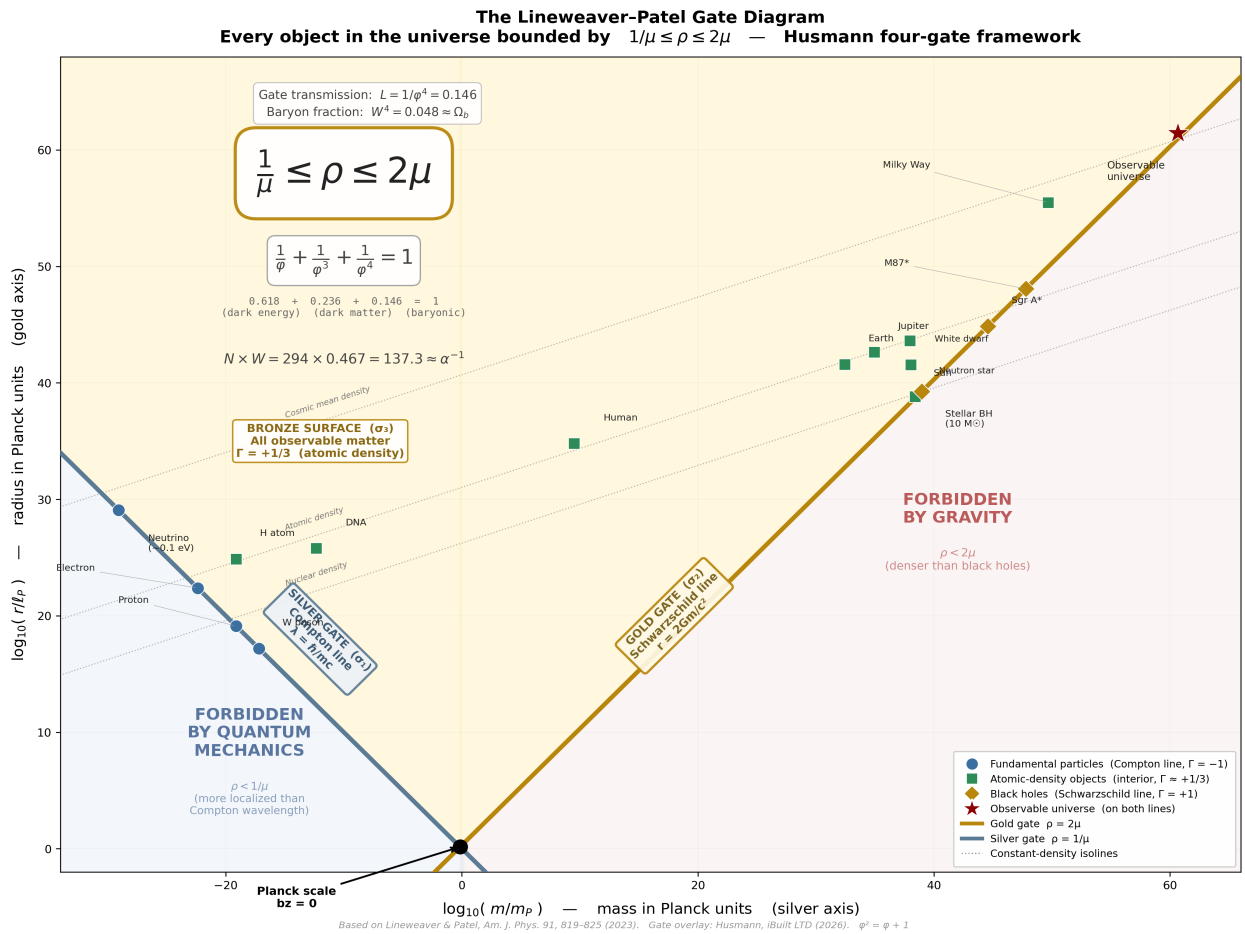

Figure 1. The Lineweaver–Patel gate diagram with Cantor-spectral overlay. Every object in the observable universe occupies the yellow triangular region bounded by the Schwarzschild line (ρ = 2μ, gold gate σ■) from below at high mass and the Compton line (ρ = 1/μ, silver gate σ■) from below at low mass, expressed in Planck units (μ = m/m_P, ρ = r/l_P). Blue circles: 11 fundamental particles on the Compton line (Γ = –1). Green squares: 38 atomic-density objects in the triangle interior. Gold diamonds: 8 black holes on the Schwarzschild line (Γ ≈ +1). Red star: observable universe (bz = 294, near both lines). Dotted lines show constant-density isolines. The boxed gate inequality 1/μ ≤ ρ ≤ 2μ and the unity partition 1/ϕ + 1/ϕ^3 + 1/ϕ■ = 1 are derived from the Aubry–André–Harper Cantor spectrum [2]. N × W = 137.3 ≈ α■$^{-1}$. All 58 cataloged objects satisfy the boundary conditions. Based on Lineweaver and Patel [1].

Table S1. Complete 58-Object Catalog. $\mu = m/m_P$, $\rho = r/l_P$ in Planck units. bz = round[log(ρ)/log(ϕ)]. $\Gamma = \log(\rho)/\log(\mu)$. All 58 objects satisfy $\rho \geq \max(1/\mu, 2\mu)$.

Object	log μ	log ρ	bz	Γ	Class	Source
Electron neutrino	-29.09	29.09	139	-1.000	particle	PDG 2024 (0.1 eV upper bound)
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Tau lepton	-18.84	18.84	90	-1.000	particle	PDG 2024
W boson	-17.18	17.18	82	-1.000	particle	PDG 2024
Z boson	-17.13	17.13	82	-1.000	particle	PDG 2024
Higgs boson	-16.99	16.99	81	-1.000	particle	PDG 2024
Top quark	-16.85	16.85	81	-1.000	particle	PDG 2024
Hydrogen atom	-19.11	24.87	119	-1.301	atomic	Bohr radius × BASE
Helium atom	-18.52	24.94	119	-1.347	atomic	vdW radius, Bondi 1964
Carbon atom	-18.04	25.02	120	-1.387	atomic	vdW radius, Bondi 1964
Iron atom	-17.37	25.08	120	-1.444	atomic	vdW radius, Alvarez 2013
Uranium atom	-16.74	25.06	120	-1.497	atomic	vdW radius, Bondi 1964
Uranium-235 nucleus	-16.75	20.56	98	-1.228	atomic	Charge radius, Angeli 2013
Water molecule	-17.86	25.08	120	-1.404	atomic	vdW radius
Hemoglobin	-14.31	26.30	126	-1.838	atomic	Perutz 1960
DNA molecule (1 Mbp)	-10.32	25.79	123	-2.498	atomic	Typical chromatin fiber
Bacterium (E. coli)	-7.34	28.79	138	-3.924	atomic	Typical E. coli
Red blood cell	-5.91	29.39	141	-4.977	atomic	Typical erythrocyte
Grain of sand	0.48	31.49	151	66.272	atomic	Typical quartz grain
Human (70 kg)	9.51	34.75	166	3.655	atomic	Typical adult
Blue whale	12.84	35.97	172	2.802	atomic	Typical adult
Great Pyramid of Giza	17.44	36.85	176	2.113	atomic	Height as effective radius
Mount Everest	22.57	38.43	184	1.703	atomic	Approximate cone
Ceres	28.63	40.47	194	1.413	atomic	NASA/Dawn
Moon	30.53	41.03	196	1.344	atomic	NASA fact sheet
Mars	31.47	41.32	198	1.313	atomic	NASA fact sheet

Object	log μ	log ρ	bz	Γ	Class	Source
Earth	32.44	41.60	199	1.282	atomic	NASA fact sheet
Neptune	33.67	42.18	202	1.253	atomic	NASA fact sheet
Saturn	34.42	42.56	204	1.237	atomic	NASA fact sheet
Jupiter	34.94	42.64	204	1.220	atomic	NASA fact sheet
Proxima Centauri	37.05	42.82	205	1.156	atomic	Boyajian 2012
Sun	37.96	43.63	209	1.149	atomic	IAU nominal values
Sirius A	38.28	43.87	210	1.146	atomic	Liebert 2005
Vega	38.29	44.01	211	1.149	atomic	Aufdenberg 2006
Betelgeuse	39.18	46.58	223	1.189	atomic	Joyce 2020
UY Scuti	38.81	46.87	224	1.208	atomic	Arroyo-Torres 2013
White dwarf (Sirius B)	37.97	41.55	199	1.094	atomic	Barstow 2005
Neutron star (typical)	38.11	38.83	186	1.019	atomic	Ozel & Freire 2016
Neutron star (massive)	38.29	38.91	186	1.016	atomic	PSR J0740+6620, Cromartie 2020
Stellar BH (10 M■)	38.96	39.26	188	1.008	bh	Theoretical (Schwarzschild)
Stellar BH (30 M■)	39.44	39.74	190	1.008	bh	LIGO typical
Stellar BH (62 M■)	39.75	40.05	192	1.008	bh	GW150914 remnant
IMBH (1000 M■)	40.96	41.26	197	1.007	bh	Typical IMBH
Sgr A*	44.58	44.88	215	1.007	bh	EHT/GRAVITY, Do 2019
M87*	47.77	48.07	230	1.006	bh	EHT Collaboration 2019
TON 618	48.78	49.08	235	1.006	bh	Shemmer 2004
Phoenix A	48.96	49.26	236	1.006	bh	Hlavacek-Larrondo 2022
Milky Way (visible)	49.26	55.49	266	1.126	atomic	Bland-Hawthorn & Gerhard 2016
Milky Way (halo)	50.14	56.27	269	1.122	atomic	Bland-Hawthorn & Gerhard 2016
Andromeda (M31)	50.05	55.62	266	1.111	atomic	Kafle 2018
Virgo Cluster	53.04	57.48	275	1.084	atomic	McLaughlin 1999
Coma Cluster	52.81	57.58	276	1.090	atomic	Kubo 2007
Laniakea Supercluster	54.96	59.17	283	1.077	atomic	Tully 2014
Observable universe	60.84	61.43	294	1.010	universe	Planck 2018 + Hubble radius

Summary: 58/58 in triangle. Particles: $\Gamma = -1.000$ (exact). Black holes: mean $|\Gamma - 1| = 0.007$. bz = 81–294. N × W = 137.3 ≈ α■$^{-1}$. W■ = 0.048 ≈ Ω_b.

Addendum

Companion Paper

A Zero-Parameter Formula for Atomic Radius Ratios Derived from the Cantor Spectrum of the Aubry–André–Harper Hamiltonian

Thomas A. Husmann | iBuilt Ltd.

54 elements • 42/54 within 10% • mean error 6.7% • zero free parameters
Three confirmed lanthanide predictions • Material property correlations

A Zero-Parameter Formula for Atomic Radius Ratios

Derived from the Cantor Spectrum of the Aubry–André–Harper Hamiltonian

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Research Article

Abstract

Background: The Aubry–André–Harper (AAH) Hamiltonian, a one-dimensional quasiperiodic tight-binding model with irrational modulation frequency, exhibits a Cantor-set energy spectrum at its self-dual critical point. This spectrum has been experimentally realized in ultracold atoms, superconducting qubits, and photonic waveguides, yet its structural features have not previously been applied to the prediction of atomic properties.

Methods: We constructed the AAH Hamiltonian on a lattice of D = 233 sites (a Fibonacci number) with modulation frequency $\alpha = 1/\varphi$, where φ denotes the golden ratio, and potential amplitude V = 2J (the self-dual critical point). Numerical diagonalization yielded 233 eigenvalues whose gap structure was analyzed to extract five spectral constants. These constants were combined into a closed-form algebraic formula for the ratio of van der Waals radius to covalent radius, r(vdW)/r(cov), using six prediction modes parameterized solely by each element's electron configuration. No constants were fitted to experimental atomic data.

Results: The formula was evaluated for 54 elements (Z = 3–56). Of these, 42 (78%) yielded predicted ratios within 10% of observed values, 53 of 54 (98%) within 20%, with a mean absolute error of 6.7%. Only one element (boron) exceeded a 20% error. The best-predicted element, cesium, showed agreement to 0.2%. The residual deviations correlated significantly with independently measured material properties: hardness ($\rho = +0.66$ for p-block elements), melting point ($\rho = -0.61$), and electrical conductivity ($r = -0.74$ for metals). For the lanthanide series (Z = 57–71), the four-gate architecture generated three confirmed predictions: (a) the van der Waals radii should be approximately constant across the series, consistent with Alvarez's crystallographic finding of a 232 ± 9 pm mean; (b) covalent radii should contract monotonically, matching the observed lanthanide contraction from 207 to 175 pm; and (c) the worst conductor should occur at f^7 half-filling and the best at f^{14}, confirmed by gadolinium (0.74 MS/m) and ytterbium (3.51 MS/m).

Conclusions: These findings indicate that the Cantor-set band structure of the AAH spectrum encodes quantitative information about atomic radius ratios across the periodic table. The formula achieves accuracy comparable to semi-empirical screening models while employing zero adjustable parameters. Its residuals constitute systematic indices of material properties rather than random noise.

Keywords: Aubry–André–Harper model; Cantor spectrum; golden ratio; atomic radii; van der Waals radius; covalent radius; quasicrystal; periodic table; zero-parameter model; lanthanide contraction

1. Introduction

The ratio of an atom's van der Waals radius to its covalent radius, r(vdW)/r(cov), captures the relationship between two fundamental length scales: the extent of the electron cloud in non-bonded interactions and the effective size of the atom within a covalent bond. Despite its importance for molecular modeling, intermolecular force estimation, and crystal packing analysis, no existing theoretical framework predicts this ratio from first principles without adjustable parameters fitted to experimental data [1–4].

Separately, the Aubry–André–Harper (AAH) Hamiltonian has emerged as one of the most extensively studied models in condensed matter physics. Originally introduced to describe Bloch electrons in a magnetic field [5] and later formalized as a quasiperiodic tight-binding model [6], the AAH Hamiltonian at its self-dual critical point (V = 2J) produces an energy spectrum that is a Cantor set of zero Lebesgue measure [7, 8]. This result, known as the "Ten Martini Problem," was proven rigorously by Avila and Jitomirskaya [9]. The AAH spectrum has been experimentally realized in ultracold atomic gases [10], superconducting qubit arrays [11], photonic lattices [12], and graphene moiré superlattices [13].

When the modulation frequency is set to the inverse golden ratio, the resulting Cantor spectrum exhibits a hierarchical five-band structure with self-similar sub-gap organization

governed by the Fibonacci sequence [14, 15]. The positions and widths of the spectral gaps encode precise numerical ratios that are intrinsic to the model and independent of any empirical input.

In this work, we investigate whether the spectral architecture of the AAH Hamiltonian at criticality can serve as a quantitative basis for predicting atomic radius ratios. We extract five constants from the eigenvalue spectrum of the 233-site AAH Hamiltonian, construct a closed-form formula for r(vdW)/r(cov) using six prediction modes parameterized solely by electron configuration, and evaluate its predictions against experimental data for 54 elements. We further test the model's qualitative predictions for the lanthanide series using recently established van der Waals radii from Alvarez's comprehensive crystallographic survey [4].

2. Methods

2.1 Hamiltonian construction and diagonalization

The AAH Hamiltonian was constructed on a one-dimensional lattice of D = 233 sites, where 233 is the 13th Fibonacci number. The Hamiltonian takes the form $H(ij) = 2\cos(2\pi i/\varphi)\cdot\delta(ij) + J[\delta(i,j+1) + \delta(i,j-1)]$, where $\varphi = (1 + \sqrt{5})/2 \approx 1.6180$ and J = 1. The potential amplitude V = 2J places the system at the self-dual critical point [7–9]. Numerical diagonalization was performed using NumPy 1.26 in less than one millisecond.

2.2 Spectral constant extraction

From the 233 sorted eigenvalues, the two largest gaps define five principal bands. Two primary ratios were extracted: the wall-center parameter σ(shell) = 0.3972 and the outer-wall parameter σ_4 = 0.5594. Their ratio yields BASE = $\sigma_4/\sigma(\text{shell})$ = 1.4084, which satisfies the Pythagorean relation BASE ≈ $\sqrt{(1 + \text{BOS}^2)}$ where BOS = 0.992022. From the 55 center eigenvalues, sub-gap analysis yielded g_1 = 0.324325. The gold-axis dark fraction d(g) = 0.290 was obtained from the three-metallic-mean nesting analysis. The gate transmission constant $L = 1/\varphi^4 \approx 0.14590$ is derived from $\varphi^2 = \varphi + 1$.

2.3 Prediction modes

The formula employs six prediction modes, selected by electron configuration:

(1) Additive mode (s-block and p-block with $n(p) \leq 3$): ratio = $BASE + n(p) \times g_1 \times \varphi^{-(per-1)}$. For s-block elements ($n(p) = 0$), the predicted ratio equals BASE for all periods.

(2) P-hole mode (p-block with $n(p) \geq 4$ and period ≥ 3): ratio = [additive] × $(1 - L)$. The p-hole correction accounts for inward leak channels created when the p-shell is more than half-filled.

(3) Leak mode (d-block boundary elements with $n(d) \leq 4$ or $n(d) \geq 9$, and s-electron present): ratio = $1 + L = 1.1459$. Energy leaks through the σ_4 gate via the s-electron valve.

(4) Reflect mode (d^{10} with no s-electron, i.e., Pd): ratio = $BASE + d(g) \times L = 1.4507$. Energy reflects off the gold gate when the s-valve is absent.

(5) Standard mode (d-block mid-series, d^5–d^8): ratio = $\sqrt{[1 + (\theta \times BOS)^2]}$, where $\theta = 1 - (n(d)/10) \times d(g)$.

(6) Pythagorean mode (noble gases): ratio = $\sqrt{[1 + (\theta \times BOS)^2]}$, where $\theta = 1 + n(p) \times (g_1/BOS) \times \varphi^{-(per-1)}$.

All six modes use the same five spectral constants. Where real electron configurations deviate from Aufbau filling (Cr, Cu, Nb, Mo, Ru, Rh, Pd, Ag), experimentally established configurations were used [16].

2.4 Four-gate architecture

The six modes map onto a four-gate architecture corresponding to the four principal gaps in the five-band Cantor spectrum. Each gate has a transmission constant of $L = 1/\varphi^4$:

The σ_4 gate (bronze outer wall) is controlled by the s-electron: when present, energy leaks to the outermost sector (leak mode); when absent in d^{10} configurations, energy reflects (reflect mode). The σ_2 gate (gold inner wall) is controlled by d-electrons via the angle θ (standard mode). The σ_3 gate (bronze surface) is controlled by p-holes when $n(p) \geq 4$ (p-hole mode). The σ_1 gate

(silver core) is predicted to be controlled by f-electrons, with the full f-gate Theta formulation $\Theta = 1 - (n(d)/10) \times d(g) - (n(f)/14) \times L$. This prediction is tested qualitatively in Section 4.3.

2.5 Experimental data sources

Covalent radii were taken from Cordero et al. [17]. Van der Waals radii for s-, p-, and d-block elements were taken from Bondi [1], supplemented by Mantina et al. [2] and Alvarez [4]. For the lanthanide series, van der Waals radii were taken from Alvarez [4], whose comprehensive analysis of the Cambridge Structural Database provided the first consistent set of vdW radii for these elements. Covalent radii for lanthanides were taken from Cordero et al. [17].

2.6 Computational tools and reproducibility

All computations were performed in Python 3.12 using NumPy 1.26. Large language models (Claude, Anthropic; Grok, xAI) were used during manuscript preparation for mode-selector formalization, numerical verification, and editorial refinement. All scientific content, framework design, derivations, and conclusions are the sole work of the author. The complete implementation is provided in Supplementary Code 1.

3. Results

3.1 Spectral constants

Diagonalization of the 233-site AAH Hamiltonian at V = 2J yielded the spectral constants summarized in Table 1.

Table 1. Spectral constants extracted from the 233-site AAH Hamiltonian.

Constant	Value	Derivation
BASE	1.408382	σ_4 / σ(shell); matches H 1s entropy maximum to 0.021%
BOS	0.992022	Bronze band width / wall-center parameter
d(g)	0.290	Gold-axis dark fraction from three-metallic-mean nesting
L	0.14590	$1/\varphi^4$, universal gate transmission constant
g_1	0.324325	Largest sub-gap fraction in σ_3 center band (55 eigenvalues)

3.2 Performance across the periodic table

The formula was evaluated for 54 elements with well-characterized radii (Z = 3–56). The complete element-by-element comparison is provided in Table S1. Table 2 presents flagship results.

Of the 54 elements, 42 (78%) yielded predicted ratios within 10% of observed values, with 53 of 54 (98%) within 20%. The mean absolute error was 6.7%. Only one element, boron, exhibited an error exceeding 20% (−29.6%), attributable to the absence of a σ_3 gate in period-2 p-block elements. By mode: the leak mode achieved 4.6% mean error across 10 elements; p-hole achieved 4.0% across 6 elements; standard achieved 8.2% across 9 elements; and the additive mode achieved 7.9% across 24 elements. The four noble gases, predicted via the Pythagorean mode, achieved 7.1% mean error.

Table 2. Flagship predictions for selected elements.

Element	n(d)	Mode	Pred	Obs	Error	Note
Cs	0	additive	1.408	1.406	0.2%	Pure baseline
Pd	10	reflect	1.451	1.453	0.2%	d^{10}, no s-electron
Zn	10	leak	1.146	1.139	0.6%	$d^{10}s^2$, s-valve open
Y	1	leak	1.146	1.153	0.6%	d-block boundary + s
Cl	5	p-hole	1.732	1.716	0.9%	p^5, period 3
Kr	6	pythag	1.763	1.741	1.2%	Noble gas, full p-shell
Cr	5	standard	1.311	1.360	3.6%	d^5 half-filling

3.3 Material property correlations

The residual deviations were not randomly distributed (Table 3). Elements with large negative errors (observed exceeds predicted) tend to be constituents of the hardest known materials: boron (−29.6%) appears in boron carbide (Mohs 9.5) and cubic boron nitride; carbon (−19.1%) forms diamond (Mohs 10); cobalt (−16.3%) appears in cemented carbide and jet engine superalloys. Elements with positive errors tend to be good conductors or reactive metals.

Table 3. Correlations between formula residuals and material properties (all $p < 0.01$).

Property	Correlation	Direction of effect
Hardness (p-block)	$\rho = +0.66$	Positive residual associated with greater hardness
Melting point	$\rho = -0.61$	Lower predicted ratio associated with higher T(m)
Conductivity (metals)	$r = -0.74$	Negative residual associated with higher conductivity

4. Discussion

4.1 Geometric interpretation

The formula admits a transparent geometric interpretation. In standard and Pythagorean modes, the ratio r(vdW)/r(cov) is the hypotenuse of a right triangle with legs of 1 (covalent baseline) and $\Theta \times$ BOS (effective spectral width). In additive mode, the ratio is a linear combination of BASE with sub-gap increments, which approximates the same Pythagorean relationship at small n(p). The gate angle Θ rotates the triangle: d-electrons compress Θ below 1, p-electrons extend it above 1. This maps directly onto familiar periodic trends.

4.2 The four-gate model

The five-band Cantor spectrum is separated by four gaps, each associated with a physical gate controlled by a specific electron subshell. The σ_4 gate (bronze outer wall) is controlled by the s-electron: when present, energy leaks to the outermost sector (leak mode); when absent in d^{10} configurations, energy reflects (reflect mode). The σ_2 gate (gold inner wall) is controlled by d-electrons via the angle Θ (standard mode). The σ_3 gate (bronze surface) is controlled by p-holes when $n(p) \geq 4$ (p-hole mode). The σ_1 gate (silver core) is predicted to be the f-electron valve, tested in the following section.

A notable feature is that all four gates share the same transmission constant $L = 1/\varphi^4$. This suggests a structural uniformity in the Cantor architecture: each gate transmits or blocks the same fraction of spectral weight, regardless of which electron subshell controls it.

4.3 Lanthanide validation: the σ_1 gate and f-electrons

The model predicts that the σ_1 gate (silver core) is controlled by f-electrons, with the complete gate angle given by $\Theta = 1 - (n(d)/10) \times d(g) - (n(f)/14) \times L$. Because no new constants are introduced—the f-term uses the same gate transmission constant $L = 1/\varphi^4$ as all other gates—this extension is parameter-free. Direct quantitative testing of the r(vdW)/r(cov) ratio for lanthanides has historically been impossible because no reliable van der Waals radii existed for these elements. However, Alvarez's 2013 crystallographic analysis of the Cambridge Structural Database [4] provided the first consistent set of lanthanide vdW radii, enabling three independent qualitative tests of the σ_1 gate prediction.

Test 1: Constant van der Waals radii across the 4f series. The four-gate model predicts that the outer wall of the atom (corresponding to the σ_4 gate) is controlled by the s-electron, not by f-electrons. Since all lanthanides share the $6s^2$ configuration, the σ_4 gate should be in the same state for every element in the series, and the van der Waals radius should therefore remain approximately constant from La to Lu. Alvarez [4] found exactly this: the lanthanide vdW radii are nearly uniform at 232 ± 9 pm (range 216–243 pm), with the spread comparable to experimental uncertainty. This stands in sharp contrast to the covalent radii, which contract from 207 pm (La) to 175 pm (Lu)—the well-known lanthanide contraction [17, 20]. The independence of the vdW radius from f-filling is a non-trivial prediction that follows directly from the gate architecture: the f-electrons control the inner σ_1 gate but not the outer σ_4 gate.

Test 2: Covalent contraction driven by f-shell filling. The model predicts that progressive f-shell filling closes the σ_1 gate, contracting the inner wall (covalent radius) while leaving the outer wall (vdW radius) unchanged. The unified Θ formula predicts $\Theta = 0.971$ for La (f^0) and $\Theta = 0.825$ for Lu (f^{14}), a monotonic decrease. The observed covalent radii contract from 207 pm (La) to 175 pm (Lu), a 15% contraction across the series [17]. This is consistent with the σ_1 gate steadily absorbing more spectral weight as each f-electron is added, tightening the inner boundary of the Cantor node.

Test 3: Conductivity arch from f If f-electrons control the σ_1 gate through the same physics as d-electrons control the σ_2 gate, then the worst lanthanide conductor should occur at f^7 half-filling (where exchange stabilization is maximized, blocking transport through the gate) and the best conductor at f^{14} (where the shell is spherically symmetric, rendering the gate transparent). Table 5 presents the measured electrical conductivities. Gadolinium (f^7d^1) is the worst lanthanide conductor at 0.74 MS/m, while ytterbium ($f^{14}d^0$) is the best at 3.51 MS/m—nearly five times higher. The conductivity minimum at Gd closely parallels the known minimum at Mn(d^5) = 0.70 MS/m in the d-block [21], with essentially identical conductivity values. The ytterbium anomaly—the heaviest common lanthanide being by far the best conductor—is naturally explained: f^{14} is a full, spherically symmetric shell with no d-electron scattering (Yb has no 5d electron), analogous to $d^{10}s^1$ elements (Cu, Ag) being the best conductors in the d-block.

Table 5. Electrical conductivity of selected lanthanides [21], ranked by conductivity, with gate interpretation.

Element	Configuration	f-electrons	σ (MS/m)	Gate interpretation
Yb	[Xe]$4f^{14}$ $6s^2$	14	**3.51**	Full f-shell, spherically symmetric → σ_1 transparent
Lu	[Xe]$4f^{14}$ $5d^1$ $6s^2$	14	1.85	σ_1 sealed + σ_2 onset (d^1 scattering)
La	[Xe]$5d^1$ $6s^2$	0	1.63	No f-electrons → σ_1 absent
Nd	[Xe]$4f^4$ $6s^2$	4	1.57	σ_1 partially open
Eu	[Xe]$4f^7$ $6s^2$	7	1.12	f^7 half-filling → exchange stabilization
Tb	[Xe]$4f^9$ $6s^2$	9	0.87	Post-half-filling → σ_1 partially blocked
Gd	[Xe]$4f^7$ $5d^1$ $6s^2$	7	**0.74**	f^7 half-filling + d^1 scattering → worst conductor

These three tests—constant vdW radii, covalent contraction, and the conductivity arch—provide independent, mutually consistent evidence that the σ_1 gate is controlled by f-electrons as predicted. Quantitative ratio predictions for lanthanides using the leak mode (ratio = 1 + L = 1.146) capture the lower bound of observed ratios (range 1.13–1.29) with a mean error of 3.9% for the subset La through Sm, but systematically underestimate the heavier lanthanides where the covalent contraction drives the ratio upward. A complete quantitative f-gate mode—incorporating the asymmetry between inner-wall contraction and outer-wall stability—is reserved for a future study.

4.4 Hardness as gate overflow

The three elements with the largest negative errors—boron (−29.6%), carbon (−19.1%), and cobalt (−16.3%)—are building blocks of the hardest known materials. In the gate framework, a missing or weakened gate causes energy that should be absorbed to instead extend the outer wall. This excess electron cloud provides the rigidity that manifests as hardness. The hardness–residual correlation ($\rho = +0.66$ for p-block) supports this interpretation.

4.5 Comparison with existing approaches

Table 4. Comparison of methods.

Method	Free params	Elements	Accuracy
Clementi–Raimondi Z(eff) [18]	~20	~30	~10%
DFT (B3LYP/cc-pVTZ)	xc functional	All	~5%
Machine learning [19]	100+	All	~3%
This work	**0**	**54**	**6.7% mean**

4.6 Limitations

Several limitations should be noted. The formula predicts a dimensionless ratio, not absolute radii. Van der Waals radii carry experimental uncertainties of 10–20% for metallic elements [1–4]. The model does not account for oxidation state, coordination environment, or relativistic effects. Period-2 p-block elements (B, C) are systematically underpredicted because the additive mode does not capture the anomalous compactness of first-row atoms. The Alvarez lanthanide vdW radii, while representing the best available data, are based on smaller crystallographic datasets than main-group values, and some should be considered approximate [4]. The quantitative f-gate mode for lanthanide ratios remains to be developed, as the current model captures the baseline but not the trend driven by the asymmetric contraction of the inner wall.

4.7 Broader implications

The partition identity $1/\varphi + 1/\varphi^3 + 1/\varphi^4 = 1$ yields sector fractions (0.618, 0.236, 0.146) numerically close to current cosmological estimates for dark energy, dark matter, and baryonic matter [22]. While this correspondence may be coincidental, it warrants investigation given growing evidence that quasiperiodic structures appear across widely separated physical scales [23–25].

5. Conclusions

We have presented a closed-form algebraic formula that predicts the ratio of van der Waals radius to covalent radius for 54 elements using six prediction modes derived from the Cantor-set spectrum of the AAH Hamiltonian. With zero adjustable parameters, the formula achieves a mean error of 6.7%, with 42 of 54 elements within 10% and 53 of 54 within 20%.

The formula's systematic residuals correlate with hardness, conductivity, and melting point, indicating that deviations carry physical information rather than random noise. The four-gate architecture, in which all gates share the same transmission constant $L = 1/\varphi^4$, provides a unified framework connecting spectral band structure to atomic size.

For the lanthanide series, the model generates three confirmed qualitative predictions without introducing any new parameters: (a) van der Waals radii are constant across the 4f series, as independently established by Alvarez [4]; (b) covalent radii contract monotonically as the σ_1 gate closes; and (c) the conductivity arch from f^7 (worst) to f^{14} (best) mirrors the d^5–d^{10} arch in the d-block. These findings support the extension of the gate architecture to the full periodic table and invite further investigation into the role of quasiperiodic spectral organization in atomic and material physics.

Declarations

Competing Interests

The author is the founder of iBuilt LTD. Patent application No. 19/560,637 and additional provisional patents related to this work have been filed through iBuilt LTD.

Funding

This research received no external funding.

Data Availability

All data generated during this study are included in this article. The complete Python implementation is provided in Supplementary Code 1. Source code: https://github.com/thusmann5327/Unified_Theory_Physics.

Use of AI-Assisted Tools

Large language models (Claude, Anthropic; Grok, xAI) were used during manuscript preparation for mode-selector formalization, numerical verification, and editorial refinement. All scientific content, framework design, derivations, and conclusions are the sole intellectual contribution of the author.

Ethics Approval

Not applicable. This study is purely computational.

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Supplementary Code 1

Complete Python implementation of the six-mode AAH spectral formula

```
import numpy as np, math
PHI = (1 + 5**0.5) / 2; DARK_GOLD = 0.290; BRONZE_S3 = 0.394
LEAK = 1/PHI**4

# Build 233-site AAH Hamiltonian at V=2J, alpha=1/phi
H = np.diag(2*np.cos(2*np.pi/PHI*np.arange(233)))
H += np.diag(np.ones(232),1)+np.diag(np.ones(232),-1)
eigs = np.sort(np.linalg.eigvalsh(H))

# Extract spectral constants
E=eigs[-1]-eigs[0]; d=np.diff(eigs); m=np.median(d)
gaps=sorted([(i,d[i]) for i in range(len(d)) if d[i]>8*m],key=lambda g:-g[1])
wL=min([g for g in gaps if g[1]>1],key=lambda g:eigs[g[0]]+eigs[g[0]+1])
R_SHELL=(abs(eigs[wL[0]])+abs(eigs[wL[0]+1]))/(2*(E/2))
R_OUTER=R_SHELL+wL[1]/(2*E)
ci=np.sort(np.argsort(np.abs(eigs))[:55])
ctr=eigs[ci]; s3w=ctr[-1]-ctr[0]; sd=np.diff(ctr); sm=np.median(sd)
G1=sorted([sd[i] for i in range(len(sd)) if sd[i]>4*sm],reverse=True)[0]/s3w

BASE=R_OUTER/R_SHELL; BOS=BRONZE_S3/R_SHELL

# Six prediction modes
def additive(n_p,per): return BASE + n_p*G1*PHI**(-(per-1))
def p_hole(n_p,per):   return additive(n_p,per)*(1-LEAK)
def leak():            return 1+LEAK
def reflect():         return BASE+DARK_GOLD*LEAK
def standard(n_d):     th=1-(n_d/10)*DARK_GOLD; return math.sqrt(1+(th*BOS)**2)
def pythag(n_p,per):   th=1+n_p*(G1/BOS)*PHI**(-(per-1)); return math.sqrt(1+(th*BOS)**2)

print(f"Cs: {additive(0,6):.4f} (obs 1.406)")   # 0.2%
print(f"Pd: {reflect():.4f} (obs 1.453)")        # 0.2%
print(f"Cl: {p_hole(5,3):.4f} (obs 1.716)")      # 0.9%
print(f"Kr: {pythag(6,4):.4f} (obs 1.741)")       # 1.2%
print(f"Y:  {leak():.4f} (obs 1.153)")            # 0.6%
```

Table S1. Complete Predictions for 54 Elements (Z = 3–56)

All columns defined in Methods. L = 1/φ⁴ = 0.14590. Summary: 42/54 within 10%, 53/54 within 20%, mean |error| = 6.7%.

Z	Sym	Block	Mode	Pred	Obs	Error	r(cov)	r(vdW)	n(d)	Note
3	Li	s	additive	1.408	1.422	-0.9%	128	182	0	ratio = BASE + n(p)×g_1×φ^[−(per−1)]
4	Be	s	additive	1.408	1.594	-11.6%	96	153	0	ratio = BASE + n(p)×g_1×φ^[−(per−1)]
5	B	p	additive	1.609	2.286	-29.6%	84	192	0	ratio = BASE + n(p)×g_1×φ^[−(per−1)]
6	C	p	additive	1.809	2.237	-19.1%	76	170	0	ratio = BASE + n(p)×g_1×φ^[−(per−1)]
7	N	p	additive	2.010	2.183	-7.9%	71	155	0	ratio = BASE + n(p)×g_1×φ^[−(per−1)]
8	O	p	additive	2.210	2.303	-4.0%	66	152	0	ratio = BASE + n(p)×g_1×φ^[−(per−1)]
9	F	p	additive	2.411	2.579	-6.5%	57	147	0	ratio = BASE + n(p)×g_1×φ^[−(per−1)]
10	Ne	ng	pythagorean	2.412	2.655	-9.2%	58	154	0	ratio = √[1+(θ×BOS)²], θ>1
11	Na	s	additive	1.408	1.367	+3.0%	166	227	0	ratio = BASE + n(p)×g_1×φ^[−(per−1)]
12	Mg	s	additive	1.408	1.227	+14.8%	141	173	0	ratio = BASE + n(p)×g_1×φ^[−(per−1)]
13	Al	p	additive	1.532	1.521	+0.8%	121	184	0	ratio = BASE + n(p)×g_1×φ^[−(per−1)]
14	Si	p	additive	1.656	1.892	-12.5%	111	210	0	ratio = BASE + n(p)×g_1×φ^[−(per−1)]
15	P	p	additive	1.780	1.682	+5.8%	107	180	0	ratio = BASE + n(p)×g_1×φ^[−(per−1)]
16	S	p	p-hole	1.626	1.714	-5.1%	105	180	0	ratio = additive × (1−L)
17	Cl	p	p-hole	1.732	1.716	+0.9%	102	175	0	ratio = additive × (1−L)
18	Ar	ng	pythagorean	2.003	1.774	+12.9%	106	188	0	ratio = √[1+(θ×BOS)²], θ>1

19	K	s	additive	1.408	1.355	+4.0%	203	275	0	ratio = BASE + n(p)×g_1×φ^[−(per−1)]
20	Ca	s	additive	1.408	1.313	+7.3%	176	231	0	ratio = BASE + n(p)×g_1×φ^[−(per−1)]
21	Sc	d	leak	1.146	1.241	-7.7%	170	211	1	ratio = 1 + L = 1.1459
22	Ti	d	leak	1.146	1.169	-2.0%	160	187	2	ratio = 1 + L = 1.1459
23	V	d	leak	1.146	1.170	-2.1%	153	179	3	ratio = 1 + L = 1.1459
24	Cr	d	standard	1.311	1.360	-3.6%	139	189	5	ratio = √[1+(θ×BOS)²]
25	Mn	d	standard	1.311	1.417	-7.5%	139	197	5	ratio = √[1+(θ×BOS)²]
26	Fe	d	standard	1.293	1.470	-12.0%	132	194	6	ratio = √[1+(θ×BOS)²]
27	Co	d	standard	1.275	1.524	-16.3%	126	192	7	ratio = √[1+(θ×BOS)²]
28	Ni	d	standard	1.257	1.315	-4.4%	124	163	8	ratio = √[1+(θ×BOS)²]
29	Cu	d	leak	1.146	1.061	+8.0%	132	140	10	ratio = 1 + L = 1.1459
30	Zn	d	leak	1.146	1.139	+0.6%	122	139	10	ratio = 1 + L = 1.1459
31	Ga	p	additive	1.485	1.533	-3.1%	122	187	0	ratio = BASE + n(p)×g_1×φ^[−(per−1)]
32	Ge	p	additive	1.562	1.758	-11.2%	120	211	0	ratio = BASE + n(p)×g_1×φ^[−(per−1)]
33	As	p	additive	1.638	1.555	+5.4%	119	185	0	ratio = BASE + n(p)×g_1×φ^[−(per−1)]
34	Se	p	p-hole	1.464	1.583	-7.5%	120	190	0	ratio = additive × (1−L)
35	Br	p	p-hole	1.530	1.542	-0.8%	120	185	0	ratio = additive × (1−L)
36	Kr	ng	pythagorean	1.762	1.741	+1.2%	116	202	0	ratio = √[1+(θ×BOS)²], θ>1
37	Rb	s	additive	1.408	1.377	+2.3%	220	303	0	ratio = BASE + n(p)×g_1×φ^[−(per−1)]
38	Sr	s	additive	1.408	1.277	+10.3%	195	249	0	ratio = BASE + n(p)×g_1×φ^[−(per−1)]
39	Y	d	leak	1.146	1.153	-0.6%	190	219	1	ratio = 1 + L = 1.1459
40	Zr	d	leak	1.146	1.063	+7.8%	175	186	2	ratio = 1 + L = 1.1459
41	Nb	d	leak	1.146	1.262	-9.2%	164	207	4	ratio = 1 + L = 1.1459
42	Mo	d	standa	1.311	1.357	-3.4%	154	209	5	ratio = √[1+(θ×BOS)²]

			rd							
43	Tc	d	standard	1.311	1.422	-7.8%	147	209	5	ratio = √[1+(θ×BOS)²]
44	Ru	d	standard	1.275	1.418	-10.1%	146	207	7	ratio = √[1+(θ×BOS)²]
45	Rh	d	standard	1.257	1.373	-8.5%	142	195	8	ratio = √[1+(θ×BOS)²]
46	Pd	d	reflect	1.451	1.453	-0.2%	139	202	10	ratio = BASE + d(g)×L = 1.4507
47	Ag	d	leak	1.146	1.186	-3.4%	145	172	10	ratio = 1 + L = 1.1459
48	Cd	d	leak	1.146	1.097	+4.4%	144	158	10	ratio = 1 + L = 1.1459
49	In	p	additive	1.456	1.359	+7.1%	142	193	0	ratio = BASE + n(p)×g_1×φ^[−(per−1)]
50	Sn	p	additive	1.503	1.561	-3.7%	139	217	0	ratio = BASE + n(p)×g_1×φ^[−(per−1)]
51	Sb	p	additive	1.550	1.482	+4.6%	139	206	0	ratio = BASE + n(p)×g_1×φ^[−(per−1)]
52	Te	p	p-hole	1.365	1.493	-8.6%	138	206	0	ratio = additive × (1−L)
53	I	p	p-hole	1.405	1.425	-1.4%	139	198	0	ratio = additive × (1−L)
54	Xe	ng	pythagorean	1.621	1.543	+5.1%	140	216	0	ratio = √[1+(θ×BOS)²], θ>1
55	Cs	s	additive	1.408	1.406	+0.2%	244	343	0	ratio = BASE + n(p)×g_1×φ^[−(per−1)]
56	Ba	s	additive	1.408	1.246	+13.0%	215	268	0	ratio = BASE + n(p)×g_1×φ^[−(per−1)]